

AI-Powered RFQ Data Extraction System – Database Design Reference

1. ITEM MASTER (Static Information)

- One entry per unique item code.
- Contains: item code, item name, standard UOM, default technical specifications (e.g., MOC, hardness, surface finish, heat treatment), and optional drawing URL.
- Rarely changes – only when product definition changes.

2. RFQ (Request for Quotation)

- Represents a single quotation request from a vendor.
- Contains: RFQ number, vendor/owner, due date, status (open, closed, awarded), and metadata (e.g., source file, AI processing status, confidence score).

3. RFQ LINE ITEMS (Dynamic Per RFQ)

- Each RFQ can have multiple line items.
- Each line item links to an item code from the Item Master.
- Stores: quantity, UOM, RFQ-specific commercial details (raw material costs, labour, taxes, freight), and technical overrides (if the hardness, finish, or MOC is different for this RFQ).

HOW IT WORKS:

- Item Master is the permanent reference library of products.
- RFQ is the container for a single vendor request.
- RFQ Line Items record what was requested in that RFQ for each item code.

Example:

RFQ#001 (Vendor: ABC Corp, September 2025) includes:

- Item VALVE-123: Quantity 500 pcs, Hardness HRC 45, Price ■3500/unit
- Item FLANGE-456: Quantity 200 pcs, Surface Finish Polished, Price ■2000/unit

If the same item VALVE-123 is requested again in RFQ#002, it gets a new line item with its new quantity and pricing without altering the original master data.

This structure is scalable because:

- It prevents duplication of item data.
- Each RFQ captures its own snapshot of prices and overrides.
- You can track item usage historically across multiple RFQs.